

Minimum Edge-Outerplanar Embeddings are Polynomial-Time Computable

Hantao Yu
Columbia University
hantao.yu@columbia.edu

Abstract

We prove that the minimum edge-outerplanarity of a planar graph can be computed in polynomial time, resolving an open problem of Bentz (2009). The proof was initially produced by GPT 5.5 Pro and then verified and polished manually.

1 Introduction

A plane embedding of a graph is called *k-edge-outerplanar* if the following *edge-peeling process* deletes all edges in at most k rounds: in each round, delete every edge lying on the outer face of the current embedding. The associated optimization problem asks for a planar embedding minimizing the number of edge-peeling rounds.

For the vertex analogue, namely ordinary k -outerplanarity, minimum- k embeddings are known to be computable in polynomial time [3]. Bentz introduced k -edge-outerplanar graphs in the study of edge-disjoint paths and multicut problems, and asked whether the analogous statement holds for k -edge-outerplanarity [2]. We answer this question affirmatively for loopless planar graphs.

Theorem 1.1 (Main theorem). *For every finite loopless planar graph G , a planar embedding of G with minimum possible edge-outerplanarity can be found in polynomial time.*

Remark 1.2. The proof below allows parallel edges. Thus it applies to all finite loopless planar multigraphs, and in particular to all finite simple planar graphs. We do not discuss loops.

We reduce the problem to the known polynomial-time problem of computing an embedding of minimum face-depth. The *depth* of a plane embedding is the maximum distance, in the face-adjacency graph, from the outer face to any other face. Bienstock and Monma studied polynomial-time algorithms for minimizing several such embedding distance measures [3]. Angelini, Di Battista, and Patrignani later gave an $O(n^4)$ -time algorithm for computing a minimum-depth embedding of an n -vertex planar graph [1].

1.1 Statement on AI use

The proof is generated using the agentic pipeline from the pipeline-math project ¹. The pipeline involves the use of GPT 5.5 Pro and Claude Opus 4.8, where GPT 5.5 Pro acts as a solver and Claude Opus 4.8 acts as a verifier. The proof is then verified by the authors, and thus the authors are solely responsible for the correctness of the proof.

¹<https://github.com/Pengbinghui/pipeline-math.git>

2 Preliminaries

Throughout this paper, we assume that all graphs are finite. Unless stated otherwise, graphs are allowed to have parallel edges but have no loops.

Let Γ be a plane embedding of a graph G . Let $F(\Gamma)$ be the set of faces of Γ , and let f_∞ be the outer face. The *face-adjacency graph* of Γ , denoted D_Γ , is the graph whose vertices are the faces of Γ , with an edge between two faces whenever they share a primal edge. We also define $F_\Gamma(e)$ to be the set of faces in $F(\Gamma)$ that are incident to e . If a primal edge e is a bridge, then it is incident with the same face on both sides; in this case $F_\Gamma(e)$ has one element, and the bridge contributes only a loop in the face-adjacency graph, and such loops are ignored for distances.

For a face $f \in F(\Gamma)$, define

$$d_\Gamma(f) := \text{dist}_{D_\Gamma}(f_\infty, f).$$

The *depth* of the embedding Γ is

$$\text{depth}(\Gamma) := \max_{f \in F(\Gamma)} d_\Gamma(f).$$

Including the outer face in this maximum is harmless, since $d_\Gamma(f_\infty) = 0$.

We note that the face-adjacency graph of any plane graph, connected or not, is connected: a generic arc from an interior point of a face f to a point of f_∞ , chosen transversal to the edges and avoiding the vertices, crosses only edges and thus traces a walk from f to f_∞ in D_Γ . Hence $d_\Gamma(f)$ is finite for every face f , and the edge-peeling process deletes every edge after finitely many rounds.

For an edge $e \in E(G)$, let $\lambda_\Gamma(e)$ be the round in which e is deleted by the edge-peeling process. Thus an edge lying on the original outer face has $\lambda_\Gamma(e) = 1$. Define

$$\text{eop}(\Gamma) := \max_{e \in E(G)} \lambda_\Gamma(e),$$

with the convention that $\text{eop}(\Gamma) = 0$ if $E(G) = \emptyset$.

Define

$$\text{OPT}_{\text{edge}}(G) := \min_{\Gamma} \text{eop}(\Gamma),$$

where the minimum is over all plane embeddings of G , including the choice of the outer face. Similarly, for a planar graph K , define

$$\text{OPT}_{\text{depth}}(K) := \min_{\Delta} \text{depth}(\Delta),$$

where the minimum is over all plane embeddings of K , including the choice of the outer face.

In the proof, we will use the following known theorem, which shows that computing a minimum-depth embedding of an n -vertex planar graph takes at most $O(n^4)$ time.

Theorem 2.1 (Angelini, Di Battista, Patrignani [1]). *Given an n -vertex planar graph K , one can compute in $O(n^4)$ time a plane embedding of K with minimum possible depth.*

3 Edge peeling in a fixed embedding

We first prove the exact relationship between edge-peeling rounds and face-depths in a fixed embedding.

Lemma 3.1 (Face reachability after edge deletions). *Let Γ be a plane embedding of a graph G , and let $S \subseteq E(G)$. Delete the edges of S , while retaining all vertices. Then the current face containing the original outer face is obtained by merging exactly those original faces that are reachable from f_∞ in the subgraph of D_Γ whose edges are dual to the primal edges in S .*

Proof. Deleting a non-bridge edge glues the two face-regions incident with that edge along the open arc formerly occupied by the edge. Deleting a bridge removes a slit lying in a single face-region and does not merge two distinct original faces. Therefore, after deleting all edges of S , the resulting face-regions are precisely the equivalence classes generated by gluing original faces across deleted non-bridge edges.

This is exactly connectivity in the subgraph of D_Γ whose edges correspond to the deleted primal edges. Possible identifications through a cutvertex do not create additional equivalences: locally around a vertex, moving from one original face sector to another requires passing across an incident edge sector, and such a passage is available precisely when the corresponding edge has been deleted. Hence the original faces merged with f_∞ are exactly the faces reachable from f_∞ through dual edges corresponding to S . $\square$

Lemma 3.2 (Fixed-embedding edge layer formula). *Let Γ be a plane embedding of a loopless graph G . Then, for every edge $e \in E(G)$,*

$$\lambda_\Gamma(e) = 1 + \min_{f \in F_\Gamma(e)} d_\Gamma(f).$$

Consequently,

$$\text{eop}(\Gamma) = \max_{e \in E(G)} \left(1 + \min_{f \in F_\Gamma(e)} d_\Gamma(f) \right).$$

Proof. Let S_r be the set of edges deleted after the first r peeling rounds, with $S_0 = \emptyset$. Let

$$B_r := \{f \in F(\Gamma) : d_\Gamma(f) \leq r\}.$$

We prove by induction on r that, after r rounds, the current outer face is obtained by merging exactly the original faces in B_r .

For $r = 0$, this is immediate, since before any edge is deleted the outer region is exactly the original outer face f_∞ , and $B_0 = \{f_\infty\}$.

Assume the statement holds after r rounds. At the beginning of round $r + 1$, an undeleted edge e lies on the current outer face if and only if at least one original face incident with e lies in the current outer region. By the induction hypothesis, this is equivalent to

$$\min_{f \in F_\Gamma(e)} d_\Gamma(f) \leq r.$$

Thus the edges deleted in round $r + 1$ are exactly the previously undeleted edges satisfying this inequality.

After deleting these edges, Lemma 3.1 shows that the new outer region consists of the original faces reachable from B_r by crossing one of the newly deleted edges. Hence it contains every face at distance at most $r + 1$ from f_∞ .

Conversely, if an edge is deleted in round $r + 1$, then it has an incident face of depth at most r . Crossing such an edge can only reach a face of depth at most $r + 1$, because adjacent faces in D_Γ have depths differing by at most one. Therefore no original face of depth greater than $r + 1$ enters the outer region after round $r + 1$.

Thus, after $r + 1$ rounds, the current outer face is obtained by merging exactly the original faces in B_{r+1} . The induction is complete.

It follows that an edge e is first deleted in the unique round $r + 1$ for which at least one incident face has depth at most r . Equivalently,

$$\lambda_\Gamma(e) = 1 + \min_{f \in F_\Gamma(e)} d_\Gamma(f).$$

Taking the maximum over all edges gives the formula for $\text{eop}(\Gamma)$. □

4 The auxiliary graph

Let G be a connected loopless planar graph with at least one edge. We construct an auxiliary graph $H = H(G)$ as follows.

For every edge $e = uv \in E(G)$, first subdivide e , replacing it by the path

$$u - s_e - v.$$

Then attach a pendant triangle to s_e . That is, add three new vertices

$$t_e, \quad a_e, \quad b_e$$

and four new edges

$$s_e t_e, \quad t_e a_e, \quad a_e b_e, \quad b_e t_e.$$

The cycle

$$C_e = t_e a_e b_e t_e$$

is called the *marker triangle* for e . The edge $s_e t_e$ is a bridge attaching this marker triangle to the subdivision vertex s_e .

The construction is linear:

$$|V(H)| = |V(G)| + 4|E(G)|, \quad |E(H)| = 6|E(G)|.$$

Moreover, if G is planar, then H is planar. If G is loopless, then H is simple even when G has parallel edges.

5 From edge layers to marker depths

We first prove that every embedding of G gives an embedding of H of the same depth as the edge-outerplanarity of the embedding of G .

Lemma 5.1. *Let G be connected and loopless, with at least one edge, and let $H = H(G)$. For every plane embedding Γ of G , there exists a plane embedding Δ of H such that*

$$\text{depth}(\Delta) = \text{eop}(\Gamma).$$

Consequently,

$$\text{OPT}_{\text{depth}}(H) \leq \text{OPT}_{\text{edge}}(G).$$

Proof. Fix a plane embedding Γ of G . For every edge $e \in E(G)$, choose an incident face $f_e \in F_\Gamma(e)$ of minimum depth:

$$d_\Gamma(f_e) = \min_{f \in F_\Gamma(e)} d_\Gamma(f).$$

Starting from Γ , subdivide each edge e by inserting s_e . This does not change the set of old faces or their distances from the outer face. Now draw the marker triangle C_e inside the chosen face f_e , attached to s_e by the bridge $s_e t_e$. Since there are only finitely many such attachments, they can be drawn pairwise disjointly inside their chosen faces. Call the resulting embedding of H Δ .

Let τ_e be the empty triangular face bounded by C_e . The other side of C_e , namely the side containing the bridge $s_e t_e$, is the old face f_e modified by the attachment. We first observe that

$$d_\Delta(f) = d_\Gamma(f) \quad \text{for every old face } f.$$

Indeed, the only edges of D_Δ that are not edges of D_Γ are the edges between a face f_e and its marker face τ_e ; since each τ_e is a leaf of D_Δ , whose unique neighbor is f_e , these new edges create no shortcuts between old faces, and the outer face is unchanged.

Hence τ_e is a leaf face one dual step farther from f_∞ than f_e . Therefore

$$d_\Delta(\tau_e) = d_\Gamma(f_e) + 1 = 1 + \min_{f \in F_\Gamma(e)} d_\Gamma(f) = \lambda_\Gamma(e),$$

where the last equality follows from Lemma 3.2. Note that this uses the fact that we chose to put the triangle gadget on the side of e with minimum face-depth in Γ .

It remains to check that no old face has depth larger than the maximum marker depth. Let f be an old face of depth $q > 0$ in Γ . Let

$$f_\infty = f_0, f_1, \dots, f_q = f$$

be a shortest path in D_Γ . The final step of this path crosses some original edge e incident with f_{q-1} and $f_q = f$. Thus

$$\min_{g \in F_\Gamma(e)} d_\Gamma(g) = q - 1.$$

By Lemma 3.2,

$$\lambda_\Gamma(e) = q.$$

So every old face-depth is witnessed by some edge-deletion layer. The only new faces in Δ are the marker faces τ_e . Therefore

$$\text{depth}(\Delta) = \max_{e \in E(G)} d_\Delta(\tau_e) = \max_{e \in E(G)} \lambda_\Gamma(e) = \text{eop}(\Gamma).$$

Taking the minimum over all embeddings Γ of G gives

$$\text{OPT}_{\text{depth}}(H) \leq \text{OPT}_{\text{edge}}(G).$$

□

6 From marker depths back to edge layers

We now prove the converse direction. The key point is that deleting marker gadgets and suppressing subdivision vertices cannot increase the relevant face distances.

Lemma 6.1 (Projection of face distances). *Let G be connected and loopless, with at least one edge, and let $H = H(G)$. Let Δ be a plane embedding of H . Let Γ be the plane embedding of G obtained from Δ by deleting every marker triangle together with its attaching bridge, and then suppressing every subdivision vertex s_e .*

For each face x of Δ , define its image $p(x)$ as follows. If x is not an empty triangular marker face, then $p(x)$ is the face of Γ containing the portion of x that remains after the deletions and suppressions. If $x = \tau_e$ is an empty triangular marker face, then set

$$p(x) = p(\tau_e) = p(\rho_e),$$

where ρ_e is the face on the bridge side of the marker triangle C_e .

Then, for every face x of Δ ,

$$d_\Gamma(p(x)) \leq d_\Delta(x).$$

Proof. Let

$$x_0, x_1, \dots, x_\ell = x$$

be a shortest path in D_Δ from the outer face x_0 of Δ to x . We show that, after deleting consecutive repetitions, the sequence

$$p(x_0), p(x_1), \dots, p(x_\ell)$$

contains a walk in D_Γ from the outer face of Γ to $p(x)$.

First consider one step $x_i x_{i+1}$ of the dual path. The faces x_i and x_{i+1} share some primal edge α of H . Every edge of H is of one of three kinds: an edge of a marker triangle, one of the two halves us_e or $s_e v$ of a subdivided edge, or an attaching bridge $s_e t_e$. We consider the three kinds in turn.

If α is an edge of a marker triangle, then the two faces incident with α are the empty marker face and the bridge-side face of that marker. These two faces have the same image under p , by definition.

If α is one of the two subdivided edges us_e or $s_e v$, then after suppressing s_e this crossing becomes either a crossing of the restored edge uv in Γ or a repetition if the two sides have merged. Thus $p(x_i)$ and $p(x_{i+1})$ are equal or adjacent in D_Γ .

If α is an attaching bridge $s_e t_e$, then α is a bridge of H , so it does not give an edge between two distinct faces in D_Δ . Hence this case cannot occur as a step of the dual path between distinct faces.

Therefore each step of the dual path in D_Δ maps to either a step or a repetition in D_Γ .

It remains to identify the starting face. If the outer face x_0 of Δ is not an empty triangular marker face, then $p(x_0)$ is the outer face of Γ . If $x_0 = \tau_e$ is an empty triangular marker face, then deleting that outer marker triangle opens its bridge-side face ρ_e directly to the exterior, so $p(x_0) = p(\rho_e)$ is again the outer face of Γ .

Thus the projected sequence gives a walk from the outer face of Γ to $p(x)$ of length at most $\ell = d_\Delta(x)$. Hence

$$d_\Gamma(p(x)) \leq d_\Delta(x). \quad \square$$

Lemma 6.2. *Let G be connected and loopless, with at least one edge, and let $H = H(G)$. For every plane embedding Δ of H , if Γ is the embedding of G obtained by deleting all marker gadgets and suppressing all subdivision vertices, then*

$$\text{eop}(\Gamma) \leq \text{depth}(\Delta).$$

Consequently,

$$\text{OPT}_{\text{edge}}(G) \leq \text{OPT}_{\text{depth}}(H).$$

Proof. Let

$$h = \text{depth}(\Delta).$$

Fix an original edge $e = uv \in E(G)$. Its marker triangle is

$$C_e = t_e a_e b_e t_e.$$

Since C_e is a simple cycle in the plane, it has two sides. Because the rest of H is attached to C_e only at the vertex t_e through the bridge $s_e t_e$, one side of C_e contains no other part of the graph. This side is an empty triangular face; call it τ_e . Let ρ_e be the face on the other side of C_e , the side containing the bridge $s_e t_e$.

There are two cases. First suppose τ_e is not the outer face of Δ . Then τ_e is a leaf in the face-adjacency graph D_Δ , and its unique neighbor is ρ_e . Hence

$$d_\Delta(\rho_e) = d_\Delta(\tau_e) - 1 \leq h - 1.$$

By Lemma 6.1,

$$d_\Gamma(p(\rho_e)) \leq d_\Delta(\rho_e) \leq h - 1.$$

After deleting the marker gadget, the face $p(\rho_e)$ is incident with the subdivided path

$$u - s_e - v.$$

After suppressing s_e , this same face is incident with the restored edge uv . Therefore the edge uv has an incident face in Γ of depth at most $h - 1$. By Lemma 3.2 we immediately get

$$\lambda_\Gamma(uv) \leq h.$$

Now suppose τ_e is the outer face of Δ . Deleting this outer marker triangle opens the bridge-side face ρ_e directly to the exterior. After suppressing s_e , the restored edge uv is incident with the outer face of Γ . Therefore

$$\lambda_\Gamma(uv) = 1 \leq h.$$

Here $h \geq 1$, because H has at least one marker triangle and hence at least two faces, and the face-adjacency graph of a plane graph is connected, so some face has positive depth.

Thus $\lambda_\Gamma(e) \leq h$ for every original edge $e \in E(G)$. Therefore

$$\text{eop}(\Gamma) \leq h = \text{depth}(\Delta).$$

Taking the minimum over all embeddings Δ of H gives

$$\text{OPT}_{\text{edge}}(G) \leq \text{OPT}_{\text{depth}}(H). \quad \square$$

Theorem 6.3 (Equality of optima). *Let G be a connected loopless planar graph with at least one edge, and let $H = H(G)$ be the auxiliary graph constructed above. Then*

$$\text{OPT}_{\text{edge}}(G) = \text{OPT}_{\text{depth}}(H).$$

Proof. Lemma 5.1 gives

$$\text{OPT}_{\text{depth}}(H) \leq \text{OPT}_{\text{edge}}(G).$$

Lemma 6.2 gives

$$\text{OPT}_{\text{edge}}(G) \leq \text{OPT}_{\text{depth}}(H).$$

Therefore the two quantities are equal. $\square$

7 Disconnected graphs

We now remove the connectedness assumption.

Lemma 7.1. *Let G be a loopless planar graph with connected components*

$$G_1, \dots, G_c.$$

Then

$$\text{OPT}_{\text{edge}}(G) = \max_{1 \leq i \leq c} \text{OPT}_{\text{edge}}(G_i).$$

Isolated vertices have value 0.

Proof. For the upper bound, take an optimal embedding Γ_i of each component G_i , and draw all components side by side, so that their outer faces lie in the common outer face of the resulting embedding Γ of G . The faces of Γ are the internal faces of the embeddings Γ_i , together with one common outer face f_∞ obtained by merging the outer faces of the Γ_i . Any path in D_{Γ_i} from the outer face of Γ_i to an internal face f of Γ_i is also a path in D_Γ from f_∞ to f , so

$$d_\Gamma(f) \leq d_{\Gamma_i}(f) \quad \text{for every internal face } f \text{ of } \Gamma_i.$$

By Lemma 3.2, applied to Γ and to each Γ_i , every edge e of G_i satisfies

$$\lambda_\Gamma(e) = 1 + \min_{f \in F_\Gamma(e)} d_\Gamma(f) \leq 1 + \min_{f \in F_{\Gamma_i}(e)} d_{\Gamma_i}(f) = \lambda_{\Gamma_i}(e).$$

Hence

$$\text{eop}(\Gamma) \leq \max_i \text{eop}(\Gamma_i) = \max_i \text{OPT}_{\text{edge}}(G_i).$$

For the lower bound, let Γ be any embedding of G . Fix a component G_i , and delete all other components from the drawing; let Γ'_i denote the induced embedding of G_i . This operation cannot increase face-distances for the remaining component: any path in D_Γ from f_∞ to a face projects to a walk in $D_{\Gamma'_i}$, because a step across an edge of G_i remains a step or becomes a repetition, while a step across an edge of another component becomes a repetition, as the two faces flanking that edge are merged when it is deleted; moreover, the outer face of Γ is contained in the outer face of Γ'_i . Hence, for every edge e of G_i , the incident faces of e in Γ'_i are the images of its incident faces in Γ , and their depths in Γ'_i are at most the corresponding depths in Γ . By Lemma 3.2, applied to Γ'_i and to Γ ,

$$\lambda_{\Gamma'_i}(e) \leq \lambda_\Gamma(e) \leq \text{eop}(\Gamma) \quad \text{for every } e \in E(G_i),$$

so $\text{eop}(\Gamma'_i) \leq \text{eop}(\Gamma)$, and therefore

$$\text{OPT}_{\text{edge}}(G_i) \leq \text{eop}(\Gamma)$$

for every i . Taking the maximum over i , and then minimizing over all embeddings Γ of G , gives

$$\max_i \text{OPT}_{\text{edge}}(G_i) \leq \text{OPT}_{\text{edge}}(G).$$

Combining the two inequalities proves the claim. $\square$

8 Polynomial-time algorithm

We now prove the main theorem.

Proof of Theorem 1.1. If $E(G) = \emptyset$, return any embedding of G and the value $k = 0$. Otherwise, compute the connected components of G . By Lemma 7.1, it suffices to solve each nontrivial connected component independently and take the maximum of the resulting values.

Thus assume first that G is connected and has at least one edge. Construct the auxiliary graph $H = H(G)$. This construction is linear:

$$|V(H)| = |V(G)| + 4|E(G)|, \quad |E(H)| = 6|E(G)|.$$

Run the polynomial-time minimum-depth embedding algorithm of Theorem 2.1 on H . Let Δ^* be a minimum-depth embedding of H . From Δ^* , delete every marker triangle together with its attaching bridge, and suppress every subdivision vertex s_e . Let Γ^* be the resulting embedding of G .

By Lemma 6.2,

$$\text{eop}(\Gamma^*) \leq \text{depth}(\Delta^*).$$

Since Δ^* is a minimum-depth embedding of H ,

$$\text{depth}(\Delta^*) = \text{OPT}_{\text{depth}}(H).$$

By Theorem 6.3,

$$\text{OPT}_{\text{depth}}(H) = \text{OPT}_{\text{edge}}(G).$$

Therefore

$$\text{eop}(\Gamma^*) \leq \text{OPT}_{\text{edge}}(G).$$

The reverse inequality holds by the definition of $\text{OPT}_{\text{edge}}(G)$. Hence

$$\text{eop}(\Gamma^*) = \text{OPT}_{\text{edge}}(G).$$

So Γ^* is an embedding of G with minimum possible edge-outerplanarity.

For disconnected G , apply the same procedure to each nontrivial connected component and place the resulting component embeddings side by side in a common outer face. By Lemma 7.1, this gives an optimal embedding of G .

The running time is polynomial because H has size linear in $|V(G)| + |E(G)|$, and the minimum-depth embedding subroutine runs in polynomial time. Using Theorem 2.1 directly gives an $O(|V(H)|^4)$ -time subroutine call for each connected component, hence polynomial time overall. $\square$

References

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